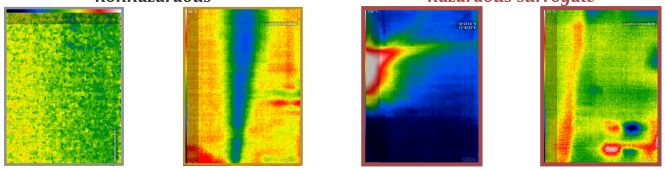
The corpus, and the action each class maps to

nonhazardous — *hazardous surrogate*

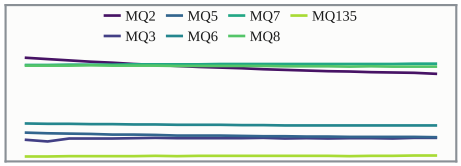


NoGas
clean air
a = 0
Monitor

Perfume
deodorant vapor
a = 1
Increase sampling

Smoke
incense smoke
a = 3
Raise alarm

Mixture
smoke and vapor
a = 4
Recommend ESD



Seven-channel MOX window, 20 × 7
the only evidence the decision component receives
the thermal path adds no term to the decision state

no training window shares a raw row with a test window

One protocol, and every comparator trained under it

Windowing
length 20
boundary windows dropped

Block-wise holdout
contiguous 20% per class, 20-window
embargo, seed-varying position

4,900 train / 1,264 test
316 windows per class

Cost-weighted policy	Unweighted network	Gradient boosting	Random forest	Recurrent (LSTM)
Conservative Q-learning	Perceptron	k-nearest neighbors	Support-vector classifier	Shallow tree depth 3

Auxiliary reference, outside the benchmark: CUSUM (two-action)

the same comparators under one protocol

In distribution	Class exclusion	Graded degradation	Raw-space faults	Closed-set reject
30 randomized runs	one hazardous class withheld	noise, drift, channel loss	stuck, zero, clamp, noise	confidence threshold
Decision accuracy	**Missed hazard**	**Escalation adequacy**	**Under-escalation**	**High-severity false alarm**

alarm burden (Eq. 9) is the high-severity false-alarm rate in operational units, not a sixth quantity